The linear transformation is now represented completely by this matrix, assuming you understand how to use it, which we shall cover. We can apply this linear transformation by juxtaposing the matrix with a vector, e.g., *Fx*.

$$F = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

We compute the result of this transformation by multiplying each row in *F* with each column (only one here) of *x*. If you do this, you get the same result as the explicit function definition above. Matrices do not need to be square, they can be any rectangular shape.

We can graphically represent matrices as boxes with two strings coming out on each end with labeled indices:



We call this a string diagram. The *n* represents the dimensionality of the input vector and the *m* is the dimensionality of the output vector. You can imagine a vector flowing into the linear transformation from the left, and a new vector is produced on the right side. For the practical deep learning we use in this book, you only need to understand this much linear algebra, i.e., the principles of multiplying vectors by matrices. Any additional math will be introduced in the respective chapters.